

V3 empirical study · Instruction

Six models · three underlyings · twelve expiry windows · 5-day hourly calibration

What this report is

- Computational source of truth: the companion Jupyter notebook. This PDF is written from the same payload; every table number is identical.
- Question: which of six American-call pricing models tracks listed market prices most closely, and does the ranking change across companies and the twelve expiry windows?
- In every table the BEST row is the lowest percentage RMSE. That row is bold and shaded green, and the Mark column ranks 1–6 (1 = lowest RMSE%).

Experimental design (held fixed for every model)

- Models: GBM, GARCH(1,1), Heston (no jumps), Merton jump-diffusion, GARCH–Merton, Heston–Merton (Bates).
- Companies: SPY (primary), AAPL, MSFT. Each name is calibrated and priced separately.
- Windows: twelve monthly third-Friday expiries (Oct 2022–Sep 2023). Evaluation is the Friday immediately before expiry, not the expiry Friday itself.
- 1-day study = that Friday. 7-day study = five NYSE sessions ending that Friday. This report is the 5-day study: the five NYSE sessions ending that Friday (35 hourly stamps; Labor Day week = 1, 5, 6, 7, 8 Sep).
- Calibration window: 5-day of 1-minute RTH equity returns and listed quotes ending at each hourly update (09:59–15:59). Parameters estimated at t use only data with timestamp $\leq t$ (no look-ahead).
- If history is shorter than the requested lookback (first hours of 10 Oct 2022), the estimator uses all available 1-minute observations up to t (equity file starts 2022-09-30).

V3 empirical study · Instruction (continued)

Six models · three underlyings · twelve expiry windows · 5-day hourly calibration

Shared market sample and filters (not model-specific)

- For each company × window, one listed-call sample is drawn once and reused by all six models: nearest-ATM call as-of each hourly evaluation timestamp (09:59, 10:59, 11:59, 12:59, 13:59, 14:59, 15:59), DTE 7–60, listed expiry.
- LSM does not run on a 15-minute or 5-minute quote grid. Friday 15:59 is listed as a scheduled stamp but `remaining_horizon = 0`, so LSM is skipped (at least two MC steps required). RMSE% uses only timestamps with a valid remaining horizon.
- Estimation filters, applied in order to every quote used for Heston/Heston–Merton calibration and to the LSM comparison set: (1) calls only, finite S , K , C ; (2) no-arbitrage $C \geq \max(0, S-K)$; (3) $7 \leq \text{DTE} \leq 60$; (4) $|S/K - 1| \leq 10\%$; (5) premium ≥ 0.05 , valid bid–ask with relative spread $\leq 50\%$ when those columns exist, volume ≥ 1 when present.
- Return-based models (GBM, Merton, GARCH, GARCH–Merton) never see option quotes in §4; they still price the same filtered LSM contracts in evaluation.

Model-correctness adjustments (V3)

- Heston / Heston–Merton: $(\kappa, \theta, \xi, \rho, v_0)$ are option-implied Fourier NLS (Albrecher little-trap), not Method A realized-variance moments. μ is the P-measure lookback mean of stock returns; LSM replaces $\mu \rightarrow r$.
- Euler ordering: the stock increment over $[t, t+\Delta t]$ uses the current variance v_t ; $v_{t+\Delta t}$ is updated afterwards. No look-ahead in the variance.
- LSM: Longstaff–Schwartz American call on the full path cloud ($n_{\text{paths}} = 10000$, seed 42). $n_{\text{steps}} =$ remaining MC steps from t to window Friday 15:59, not listed DTE. $\Delta t = 5$ minute(s). Paths are not averaged before stopping. Do not extend paths to listed expiry.

Metrics

- Primary: $\text{RMSE\%} = 100 \times \sqrt{\text{mean}(((C_{\text{model}} - C_{\text{mkt}}) / C_{\text{mkt}})^2)}$. Ranking uses this number only.
- Secondary: $\text{MAE} = \text{mean } |C_{\text{model}} - C_{\text{mkt}}|$ (dollars). $\text{Bias \$} = \text{mean}(C_{\text{model}} - C_{\text{mkt}})$ (dollars). $\text{Bias\%} = 100 \times \text{mean}((C_{\text{model}} - C_{\text{mkt}})/C_{\text{mkt}})$. Positive bias = model expensive vs market. Early-exercise fraction = mean share of LSM paths that stop before the window

Page 3 · Filters, sample, and no-look-ahead

Shared evaluation sample · n listed calls (identical for all six models)

Window	Eval Friday	Sessions	SPY n	AAPL n	MSFT n
Oct 2022 2022-10-21	2022-10-14	5	34	34	34
Nov 2022 2022-11-18	2022-11-11	5	34	34	34
Dec 2022 2022-12-16	2022-12-09	5	34	34	34
Jan 2023 2023-01-20	2023-01-13	5	34	34	34
Feb 2023 2023-02-17	2023-02-10	5	34	34	34
Mar 2023 2023-03-17	2023-03-10	5	34	34	34
Apr 2023 2023-04-21	2023-04-14	5	34	34	34
May 2023 2023-05-19	2023-05-12	5	34	34	34
Jun 2023 2023-06-16	2023-06-09	5	34	34	34
Jul 2023 2023-07-21	2023-07-14	5	34	34	34
Aug 2023 2023-08-18	2023-08-11	5	34	34	34
Sep 2023 2023-09-15	2023-09-08	5	34	34	34

Filter funnel · last-step n in the 5-day lookback ending at each window (SPY processed calls)

Window	Input*	No-arb	DTE 7-60	$ S/K-1 \leq 10\%$	Liquidity	Eval n
Oct22	5108	5108	5108	5068	5068	34
Nov22	5565	5562	5562	5476	5476	34
Dec22	4199	4164	4164	4085	4085	34
Jan23	3747	3747	3747	3710	3710	34
Feb23	4384	4281	4281	4229	4229	34
Mar23	3983	3983	3983	3942	3942	34
Apr23	3536	3536	3536	3479	3479	34
May23	3623	3623	3623	3581	3581	34
Jun23	3381	3381	3381	3313	3313	34
Jul23	3610	3610	3610	3530	3530	34
Aug23	4262	4262	4262	4183	4183	34
Sep23	3569	3569	3569	3534	3534	34

*Input is the processed call panel restricted to the 5-day 1-minute lookback ending at window Friday 15:59. Same filters for AAPL/MSFT. No look-ahead: hourly parameters at t use only data $\leq t$. LSM horizon

Page 4 · Table block · SPY · six models × four expiry windows

Same contracts and LSM settings for every model. Bold green row = lowest RMSE%. 5-day lookback, hourly rolling, $\Delta t = 5$ min, $n_{\text{paths}}=10000$, $\text{seed}=42$. LSM horizon = remaining time to

Table SPY-1 SPY · Oct 2022 listed expiry 2022-10-21
2022-10-10 09:30 → 2022-10-14 15:59 · n = 34 contracts

Model	RMSE%	MAE%	Bias%	Ranking
MD-GBM	74.58	70.42	-70.42	4
GBM	74.01	69.95	-69.95	3
GARCH	32.58	25.67	+2.76	1
Merton	262.87	235.20	+233.82	5
GARCH-Merton	72.90	68.67	-68.67	2

Table SPY-2 SPY · Nov 2022 listed expiry 2022-11-18
2022-11-07 09:30 → 2022-11-11 15:59 · n = 34 contracts

Model	RMSE%	MAE%	Bias%	Ranking
MD-GBM	99.87	99.87	-99.87	3
GBM	99.67	99.67	-99.67	2
GARCH	69.40	57.61	-57.31	1
Merton	234.03	196.04	+174.09	5
GARCH-Merton	99.93	99.93	-99.93	4

Table SPY-3 SPY · Dec 2022 listed expiry 2022-12-16
2022-12-05 09:30 → 2022-12-09 15:59 · n = 34 contracts

Model	RMSE%	MAE%	Bias%	Ranking
MD-GBM	100.00	100.00	-100.00	3
GBM	100.00	100.00	-100.00	4
GARCH	88.40	84.60	-82.66	1
Merton	192.37	159.55	+147.44	5
GARCH-Merton	100.00	100.00	-100.00	2

Table SPY-4 SPY · Jan 2023 listed expiry 2023-01-20
2023-01-09 09:30 → 2023-01-13 15:59 · n = 34 contracts

Model	RMSE%	MAE%	Bias%	Ranking
MD-GBM	100.00	100.00	-100.00	2
GBM	100.00	100.00	-100.00	3
GARCH	95.03	93.82	-93.82	1
Merton	163.83	134.29	+104.06	5
GARCH-Merton	100.00	100.00	-100.00	4

RMSE% is $100 \times \sqrt{\text{mean}(((\text{model} - \text{market}) / \text{market})^2)}$. MAE and Bias \$ are in option-premium dollars. Bias% is $100 \times \text{mean}((\text{model} - \text{market}) / \text{market})$. Early-ex. frac. is the mean LSM share of paths exercised by

Page 5 · Table block · SPY · six models × four expiry windows

Same contracts and LSM settings for every model. Bold green row = lowest RMSE%. 5-day lookahead, hourly rolling, $\Delta t = 5$ min, $n_{\text{paths}}=10000$, $\text{seed}=42$. LSM horizon = remaining time to

Table SPY-5 SPY · Feb 2023 listed expiry 2023-02-17
2023-02-06 09:30 → 2023-02-10 15:59 · n = 34 contracts

Model	RMSE%	MAE%	Bias%	Ranking
MD-GBM	100.00	100.00	-100.00	3
GBM	100.00	100.00	-100.00	2
GARCH	83.79	78.09	-76.87	1
Merton	179.75	154.58	+135.69	5
GARCH-Merton	100.00	100.00	-100.00	4

Table SPY-6 SPY · Mar 2023 listed expiry 2023-03-17
2023-03-06 09:30 → 2023-03-10 15:59 · n = 34 contracts

Model	RMSE%	MAE%	Bias%	Ranking
MD-GBM	96.11	95.72	-95.72	3
GBM	95.77	95.37	-95.37	2
GARCH	118.20	96.44	+56.21	4
Merton	299.25	264.81	+262.24	5
GARCH-Merton	95.75	95.34	-95.34	1

Table SPY-7 SPY · Apr 2023 listed expiry 2023-04-21
2023-04-10 09:30 → 2023-04-14 15:59 · n = 34 contracts

Model	RMSE%	MAE%	Bias%	Ranking
MD-GBM	100.00	100.00	-100.00	4
GBM	100.00	100.00	-100.00	3
GARCH	85.87	78.03	-5.16	1
Merton	209.44	178.03	+167.30	5
GARCH-Merton	99.93	99.93	-99.93	2

Table SPY-8 SPY · May 2023 listed expiry 2023-05-19
2023-05-08 09:30 → 2023-05-12 15:59 · n = 34 contracts

Model	RMSE%	MAE%	Bias%	Ranking
MD-GBM	100.00	100.00	-100.00	2
GBM	100.00	100.00	-100.00	1
GARCH	102.07	95.24	-2.07	4
Merton	197.37	163.95	+146.51	5
GARCH-Merton	100.00	100.00	-100.00	3

RMSE% is $100 \times \sqrt{\text{mean}(((\text{model} - \text{market}) / \text{market})^2)}$. MAE and Bias \$ are in option-premium dollars. Bias% is $100 \times \text{mean}((\text{model} - \text{market}) / \text{market})$. Early-ex. frac. is the mean LSM share of paths exercised b

Page 6 · Table block · SPY · six models × four expiry windows

Same contracts and LSM settings for every model. Bold green row = lowest RMSE%. 5-day lookahead, hourly rolling, $\Delta t = 5$ min, $n_{\text{paths}}=10000$, $\text{seed}=42$. LSM horizon = remaining time to

Table SPY-9 SPY · Jun 2023 listed expiry 2023-06-16
2023-06-05 09:30 → 2023-06-09 15:59 · n = 34 contracts

Model	RMSE%	MAE%	Bias%	Ranking
MD-GBM	100.00	100.00	-100.00	1
GBM	100.00	100.00	-100.00	2
GARCH	101.38	90.80	-21.76	4
Merton	236.00	189.92	+164.48	5
GARCH-Merton	100.00	100.00	-100.00	3

Table SPY-10 SPY · Jul 2023 listed expiry 2023-07-21
2023-07-10 09:30 → 2023-07-14 15:59 · n = 34 contracts

Model	RMSE%	MAE%	Bias%	Ranking
MD-GBM	100.00	100.00	-100.00	1
GBM	100.00	100.00	-100.00	2
GARCH	128.91	113.38	+24.53	4
Merton	241.60	197.57	+180.20	5
GARCH-Merton	100.00	100.00	-100.00	3

Table SPY-11 SPY · Aug 2023 listed expiry 2023-08-18
2023-08-07 09:30 → 2023-08-11 15:59 · n = 34 contracts

Model	RMSE%	MAE%	Bias%	Ranking
MD-GBM	100.00	100.00	-100.00	2
GBM	100.00	100.00	-100.00	3
GARCH	78.24	70.81	-57.65	1
Merton	159.06	134.25	+105.80	5
GARCH-Merton	100.00	100.00	-100.00	4

Table SPY-12 SPY · Sep 2023 listed expiry 2023-09-15
2023-09-01 09:30 → 2023-09-08 15:59 · n = 34 contracts

Model	RMSE%	MAE%	Bias%	Ranking
MD-GBM	100.00	100.00	-100.00	2
GBM	100.00	100.00	-100.00	3
GARCH	92.28	89.47	-86.79	1
Merton	142.96	113.83	+75.41	5
GARCH-Merton	100.00	100.00	-100.00	4

RMSE% is $100 \times \sqrt{\text{mean}(((\text{model} - \text{market}) / \text{market})^2)}$. MAE and Bias \$ are in option-premium dollars. Bias% is $100 \times \text{mean}((\text{model} - \text{market}) / \text{market})$. Early-ex. frac. is the mean LSM share of paths exercised b

Page 7 · Table block · AAPL · six models × four expiry windows

Same contracts and LSM settings for every model. Bold green row = lowest RMSE%. 5-day lookback, hourly rolling, $\Delta t = 5$ min, $n_{\text{paths}}=10000$, $\text{seed}=42$. LSM horizon = remaining time to

**Table AAPL-1 AAPL · Oct 2022 listed expiry 2022-10-21
2022-10-10 09:30 → 2022-10-14 15:59 · n = 34 contracts**

Model	RMSE%	MAE%	Bias%	Ranking
MD-GBM	61.31	58.72	-48.93	4
GBM	61.05	58.50	-48.72	3
GARCH	41.01	32.58	+5.91	1
Merton	254.24	236.24	+235.03	5
GARCH-Merton	60.52	58.06	-48.30	2

**Table AAPL-2 AAPL · Nov 2022 listed expiry 2022-11-18
2022-11-07 09:30 → 2022-11-11 15:59 · n = 34 contracts**

Model	RMSE%	MAE%	Bias%	Ranking
MD-GBM	96.80	96.46	-96.46	3
GBM	95.89	95.55	-95.55	2
GARCH	68.55	60.47	-60.47	1
Merton	232.52	208.27	+202.64	5
GARCH-Merton	97.12	96.67	-96.67	4

**Table AAPL-3 AAPL · Dec 2022 listed expiry 2022-12-16
2022-12-05 09:30 → 2022-12-09 15:59 · n = 34 contracts**

Model	RMSE%	MAE%	Bias%	Ranking
MD-GBM	92.26	91.75	-91.75	4
GBM	91.32	90.79	-90.79	3
GARCH	56.90	50.48	-23.51	1
Merton	254.50	231.07	+230.37	5
GARCH-Merton	88.90	88.30	-88.30	2

**Table AAPL-4 AAPL · Jan 2023 listed expiry 2023-01-20
2023-01-09 09:30 → 2023-01-13 15:59 · n = 34 contracts**

Model	RMSE%	MAE%	Bias%	Ranking
MD-GBM	99.72	99.72	-99.72	4
GBM	99.51	99.50	-99.50	3
GARCH	80.43	74.56	-72.30	1
Merton	195.11	162.93	+152.88	5
GARCH-Merton	99.42	99.42	-99.42	2

RMSE% is $100 \times \sqrt{\text{mean}(((\text{model} - \text{market}) / \text{market})^2)}$. MAE and Bias \$ are in option-premium dollars. Bias% is $100 \times \text{mean}((\text{model} - \text{market}) / \text{market})$. Early-ex. frac. is the mean LSM share of paths exercised b

Page 8 · Table block · AAPL · six models × four expiry windows

Same contracts and LSM settings for every model. Bold green row = lowest RMSE%. 5-day lookahead, hourly rolling, Δt = 5 min, n_paths=10000, seed=42. LSM horizon = remaining time to

Table AAPL-5 AAPL · Feb 2023 listed expiry 2023-02-17
2023-02-06 09:30 → 2023-02-10 15:59 · n = 34 contracts

Model	RMSE%	MAE%	Bias%	Ranking
MD-GBM	97.63	97.49	-97.49	3
GBM	96.90	96.71	-96.71	2
GARCH	65.42	56.72	-37.23	1
Merton	278.14	242.18	+238.86	5
GARCH-Merton	98.46	98.32	-98.32	4

Table AAPL-6 AAPL · Mar 2023 listed expiry 2023-03-17
2023-03-06 09:30 → 2023-03-10 15:59 · n = 34 contracts

Model	RMSE%	MAE%	Bias%	Ranking
MD-GBM	77.99	75.74	-75.74	3
GBM	77.24	75.01	-75.01	2
GARCH	125.14	104.27	+69.59	4
Merton	362.53	319.29	+317.63	5
GARCH-Merton	77.02	74.55	-74.55	1

Table AAPL-7 AAPL · Apr 2023 listed expiry 2023-04-21
2023-04-10 09:30 → 2023-04-14 15:59 · n = 34 contracts

Model	RMSE%	MAE%	Bias%	Ranking
MD-GBM	84.91	82.77	-82.77	4
GBM	84.45	82.35	-82.35	3
GARCH	63.99	52.65	+21.45	1
Merton	204.41	182.41	+178.16	5
GARCH-Merton	83.84	81.82	-81.82	2

Table AAPL-8 AAPL · May 2023 listed expiry 2023-05-19
2023-05-08 09:30 → 2023-05-12 15:59 · n = 34 contracts

Model	RMSE%	MAE%	Bias%	Ranking
MD-GBM	99.47	99.47	-99.47	2
GBM	99.19	99.17	-99.17	1
GARCH	101.12	96.96	-11.92	4
Merton	283.02	227.63	+211.22	5
GARCH-Merton	99.70	99.69	-99.69	3

RMSE% is 100×√mean(((model−market)/market)²). MAE and Bias \$ are in option-premium dollars. Bias% is 100×mean((model−market)/market). Early-ex. frac. is the mean LSM share of paths exercised b

Page 9 · Table block · AAPL · six models × four expiry windows

Same contracts and LSM settings for every model. Bold green row = lowest RMSE%. 5-day lookahead, hourly rolling, Δt = 5 min, n_paths=10000, seed=42. LSM horizon = remaining time to

Table AAPL-9 AAPL · Jun 2023 listed expiry 2023-06-16
2023-06-05 09:30 → 2023-06-09 15:59 · n = 34 contracts

Model	RMSE%	MAE%	Bias%	Ranking
MD-GBM	100.00	100.00	-100.00	4
GBM	100.00	100.00	-100.00	3
GARCH	87.18	82.61	-76.82	1
Merton	174.05	153.86	+146.86	5
GARCH-Merton	99.97	99.97	-99.97	2

Table AAPL-10 AAPL · Jul 2023 listed expiry 2023-07-21
2023-07-10 09:30 → 2023-07-14 15:59 · n = 34 contracts

Model	RMSE%	MAE%	Bias%	Ranking
MD-GBM	100.00	100.00	-100.00	3
GBM	100.00	100.00	-100.00	4
GARCH	82.42	75.42	-66.48	1
Merton	254.55	209.82	+190.89	5
GARCH-Merton	100.00	100.00	-100.00	2

Table AAPL-11 AAPL · Aug 2023 listed expiry 2023-08-18
2023-08-07 09:30 → 2023-08-11 15:59 · n = 34 contracts

Model	RMSE%	MAE%	Bias%	Ranking
MD-GBM	100.00	100.00	-100.00	3
GBM	100.00	100.00	-100.00	4
GARCH	92.85	91.41	-91.41	1
Merton	228.80	193.95	+179.22	5
GARCH-Merton	100.00	100.00	-100.00	2

Table AAPL-12 AAPL · Sep 2023 listed expiry 2023-09-15
2023-09-01 09:30 → 2023-09-08 15:59 · n = 34 contracts

Model	RMSE%	MAE%	Bias%	Ranking
MD-GBM	100.00	100.00	-100.00	3
GBM	100.00	100.00	-100.00	4
GARCH	99.76	99.75	-99.75	1
Merton	171.18	133.20	+126.40	5
GARCH-Merton	100.00	100.00	-100.00	2

RMSE% is 100×√mean(((model−market)/market)²). MAE and Bias \$ are in option-premium dollars. Bias% is 100×mean((model−market)/market). Early-ex. frac. is the mean LSM share of paths exercised b

Page 10 · Table block · MSFT · six models × four expiry windows

Same contracts and LSM settings for every model. Bold green row = lowest RMSE%. 5-day lookback, hourly rolling, $\Delta t = 5$ min, $n_{\text{paths}}=10000$, $\text{seed}=42$. LSM horizon = remaining time to

**Table MSFT-1 MSFT · Oct 2022 listed expiry 2022-10-21
2022-10-10 09:30 → 2022-10-14 15:59 · n = 34 contracts**

Model	RMSE%	MAE%	Bias%	Ranking
MD-GBM	70.77	64.78	-37.92	3
GBM	70.40	64.50	-37.59	2
GARCH	52.40	36.58	+15.03	1
Merton	237.29	221.60	+221.60	5
GARCH-Merton	71.16	65.16	-38.26	4

**Table MSFT-2 MSFT · Nov 2022 listed expiry 2022-11-18
2022-11-07 09:30 → 2022-11-11 15:59 · n = 34 contracts**

Model	RMSE%	MAE%	Bias%	Ranking
MD-GBM	62.79	50.51	-15.38	3
GBM	62.41	50.04	-14.98	2
GARCH	174.12	136.24	+100.90	4
Merton	480.15	402.44	+401.10	5
GARCH-Merton	62.35	50.10	-16.77	1

**Table MSFT-3 MSFT · Dec 2022 listed expiry 2022-12-16
2022-12-05 09:30 → 2022-12-09 15:59 · n = 34 contracts**

Model	RMSE%	MAE%	Bias%	Ranking
MD-GBM	58.83	51.60	-51.60	3
GBM	57.59	50.63	-50.63	2
GARCH	56.21	44.10	+42.87	1
Merton	288.10	273.25	+273.25	5
GARCH-Merton	60.28	52.51	-52.51	4

**Table MSFT-4 MSFT · Jan 2023 listed expiry 2023-01-20
2023-01-09 09:30 → 2023-01-13 15:59 · n = 34 contracts**

Model	RMSE%	MAE%	Bias%	Ranking
MD-GBM	27.87	24.04	+3.55	2
GBM	27.91	24.09	+3.70	3
GARCH	131.19	104.69	+103.49	4
Merton	323.61	289.34	+289.34	5
GARCH-Merton	27.16	23.32	+2.60	1

RMSE% is $100 \times \sqrt{\text{mean}(((\text{model} - \text{market}) / \text{market})^2)}$. MAE and Bias \$ are in option-premium dollars. Bias% is $100 \times \text{mean}((\text{model} - \text{market}) / \text{market})$. Early-ex. frac. is the mean LSM share of paths exercised before

Page 11 · Table block · MSFT · six models × four expiry windows

Same contracts and LSM settings for every model. Bold green row = lowest RMSE%. 5-day lookback, hourly rolling, $\Delta t = 5$ min, $n_{\text{paths}}=10000$, $\text{seed}=42$. LSM horizon = remaining time to

**Table MSFT-5 MSFT · Feb 2023 listed expiry 2023-02-17
2023-02-06 09:30 → 2023-02-10 15:59 · n = 34 contracts**

Model	RMSE%	MAE%	Bias%	Ranking
MD-GBM	96.79	89.73	-12.88	2
GBM	96.45	89.21	-12.30	1
GARCH	242.03	198.18	+133.45	4
Merton	530.35	461.60	+459.63	5
GARCH-Merton	97.16	90.30	-15.98	3

**Table MSFT-6 MSFT · Mar 2023 listed expiry 2023-03-17
2023-03-06 09:30 → 2023-03-10 15:59 · n = 34 contracts**

Model	RMSE%	MAE%	Bias%	Ranking
MD-GBM	90.61	88.87	-88.87	2
GBM	89.94	88.14	-88.14	1
GARCH	144.78	112.36	+66.65	4
Merton	479.17	419.05	+418.94	5
GARCH-Merton	91.79	90.28	-90.28	3

**Table MSFT-7 MSFT · Apr 2023 listed expiry 2023-04-21
2023-04-10 09:30 → 2023-04-14 15:59 · n = 34 contracts**

Model	RMSE%	MAE%	Bias%	Ranking
MD-GBM	98.83	98.78	-98.78	3
GBM	98.57	98.51	-98.51	2
GARCH	51.49	40.24	-25.84	1
Merton	328.01	289.01	+284.97	5
GARCH-Merton	99.29	99.23	-99.23	4

**Table MSFT-8 MSFT · May 2023 listed expiry 2023-05-19
2023-05-08 09:30 → 2023-05-12 15:59 · n = 34 contracts**

Model	RMSE%	MAE%	Bias%	Ranking
MD-GBM	100.00	100.00	-100.00	2
GBM	100.00	100.00	-100.00	3
GARCH	90.00	88.06	-88.06	1
Merton	181.23	139.98	+115.85	5
GARCH-Merton	100.00	100.00	-100.00	4

RMSE% is $100 \times \sqrt{\text{mean}(((\text{model} - \text{market}) / \text{market})^2)}$. MAE and Bias \$ are in option-premium dollars. Bias% is $100 \times \text{mean}((\text{model} - \text{market}) / \text{market})$. Early-ex. frac. is the mean LSM share of paths exercised b

Same contracts and LSM settings for every model. Bold green row = lowest RMSE%. 5-day lookback, hourly rolling, Δt = 5 min, n_paths=10000, seed=42. LSM horizon = remaining time to

Table MSFT-9 MSFT · Jun 2023 listed expiry 2023-06-16
2023-06-05 09:30 → 2023-06-09 15:59 · n = 34 contracts

Model	RMSE%	MAE%	Bias%	Ranking
MD-GBM	100.00	100.00	-100.00	3
GBM	100.00	100.00	-100.00	4
GARCH	97.45	97.30	-97.30	1
Merton	186.81	158.69	+143.94	5
GARCH-Merton	100.00	100.00	-100.00	2

Table MSFT-10 MSFT · Jul 2023 listed expiry 2023-07-21
2023-07-10 09:30 → 2023-07-14 15:59 · n = 34 contracts

Model	RMSE%	MAE%	Bias%	Ranking
MD-GBM	100.00	100.00	-100.00	2
GBM	100.00	100.00	-100.00	3
GARCH	95.23	94.82	-94.82	1
Merton	225.67	194.21	+179.51	5
GARCH-Merton	100.00	100.00	-100.00	4

Table MSFT-11 MSFT · Aug 2023 listed expiry 2023-08-18
2023-08-07 09:30 → 2023-08-11 15:59 · n = 34 contracts

Model	RMSE%	MAE%	Bias%	Ranking
MD-GBM	100.00	100.00	-100.00	2
GBM	100.00	100.00	-100.00	3
GARCH	99.99	99.99	-99.99	1
Merton	168.29	149.10	+129.04	5
GARCH-Merton	100.00	100.00	-100.00	4

Table MSFT-12 MSFT · Sep 2023 listed expiry 2023-09-15
2023-09-01 09:30 → 2023-09-08 15:59 · n = 34 contracts

Model	RMSE%	MAE%	Bias%	Ranking
MD-GBM	100.00	100.00	-100.00	2
GBM	100.00	100.00	-100.00	3
GARCH	100.00	100.00	-100.00	1
Merton	155.23	123.61	+84.65	5
GARCH-Merton	100.00	100.00	-100.00	4

Summary tables · RMSE% across companies and twelve windows

Table S1 · Overall ranking (mean RMSE% over 36 cells)

Rank	Model	Mean RMSE%	Median RMSE%	# of 36 cells best
1	GBM	91.06	99.84	4
2	GARCH-Merton	91.18	99.93	4
3	MD-GBM	91.31	99.93	2
4	GARCH	93.77	91.14	26
5	Merton	249.87	233.27	0

Table S2 · Wins by company (lowest RMSE% in that company×window)

Model	SPY wins	AAPL wins	MSFT wins	Total
MD-GBM	2	0	0	2
GBM	1	1	2	4
GARCH	8	10	8	26
Merton	0	0	0	0
GARCH-Merton	1	1	2	4

Table S3 · Mean RMSE% by window (equal-weight mean of SPY / AAPL / MSFT)

Model	Oct22	Nov22	Dec22	Jan23	Feb23	Mar23	Apr23	May23	Jun23	Jul23	Aug23	Sep23	Mean of 12
MD-GBM	68.89	86.49	83.70	75.86	98.14	88.24	94.58	99.82	100.00	100.00	100.00	100.00	91.31
GBM	68.49	85.99	82.97	75.81	97.78	87.65	94.34	99.73	100.00	100.00	100.00	100.00	91.06
GARCH	42.00	104.03	67.17	102.21	130.41	129.37	67.11	97.73	95.34	102.19	90.36	97.34	93.77
Merton	251.46	315.56	244.99	227.52	329.41	380.32	247.29	220.54	198.95	240.61	185.38	156.46	249.87
GARCH-Merton	68.19	86.47	83.06	75.53	98.54	88.19	94.35	99.90	99.99	100.00	100.00	100.00	91.18

Green row in S1 / S3 = lowest mean RMSE%. Green row in S2 = most company×window wins. A model can win the most cells without having the lowest average error (and vice versa).

Summary by company · mean RMSE% across twelve windows

Table S4.1 · SPY

Model	Mean RMSE%	Median # windows best	
MD-GBM	97.55	100.00	2
GBM	97.45	100.00	1
GARCH	89.68	90.34	8
Merton	209.88	203.40	0
GARCH-Merton	97.38	100.00	1

Table S4.2 · AAPL

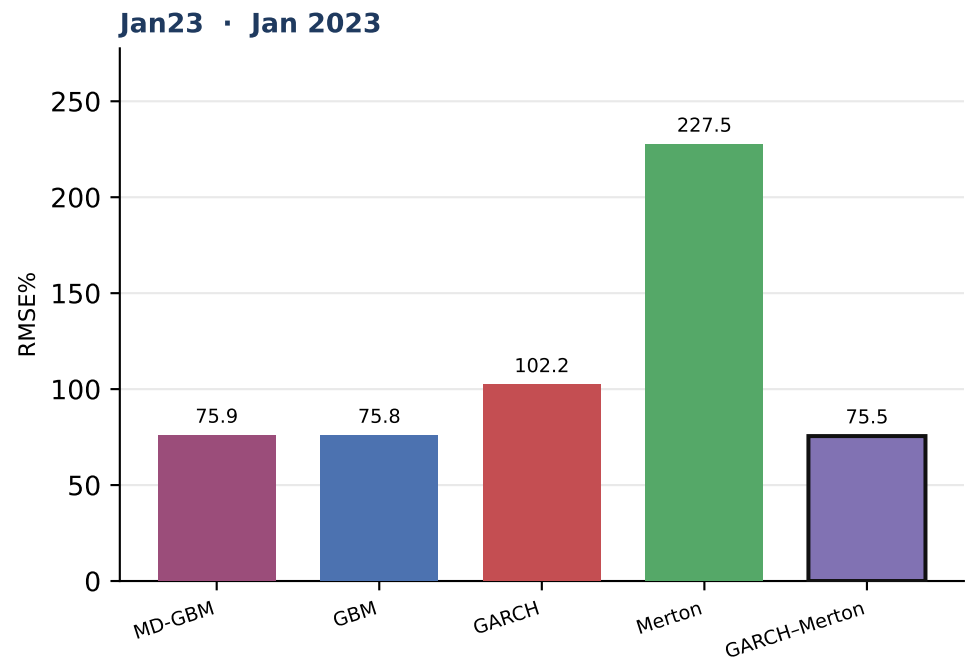
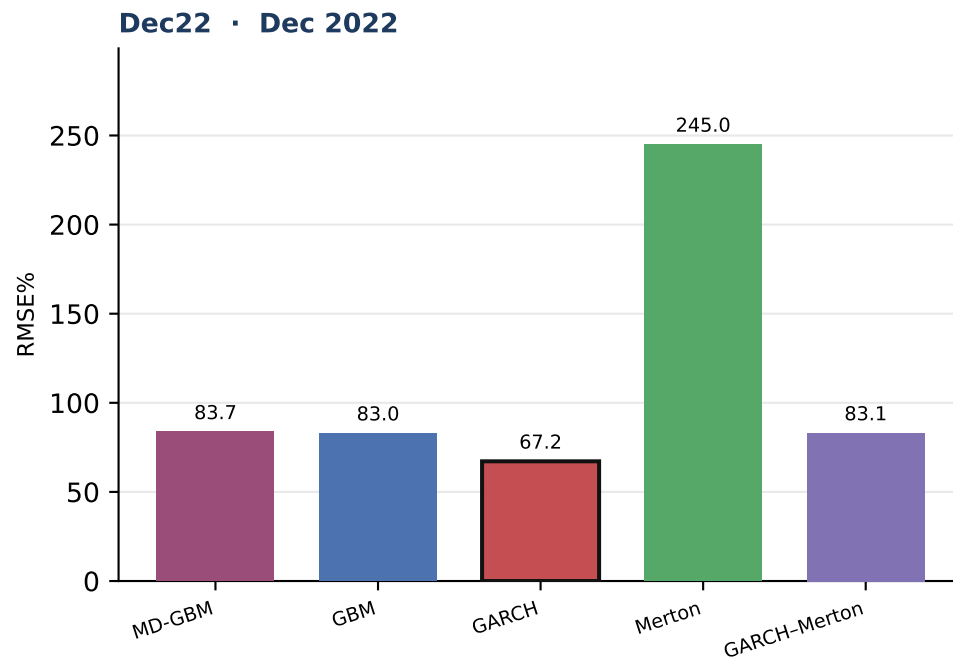
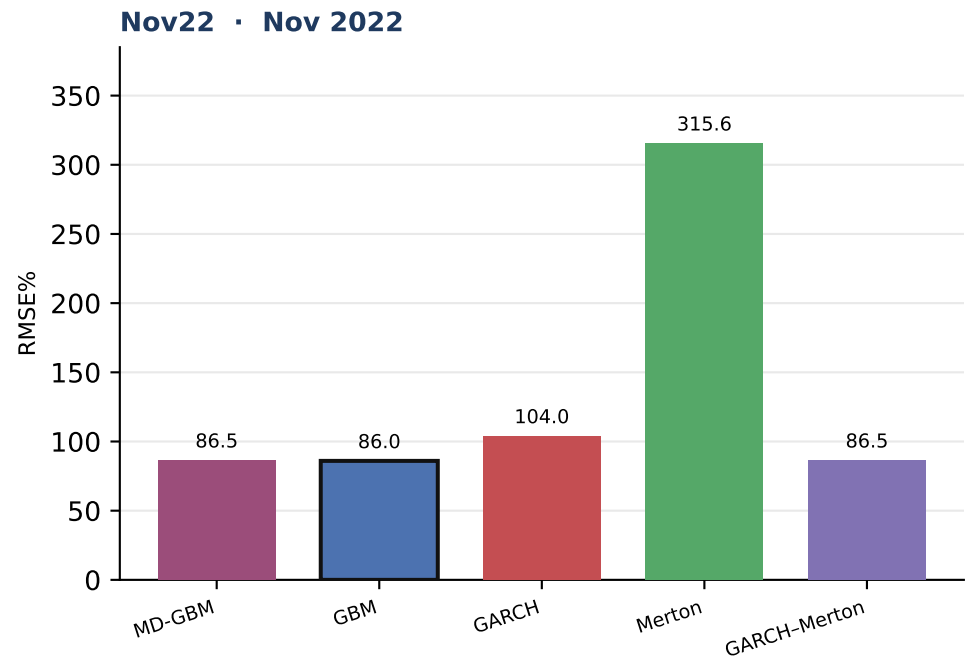
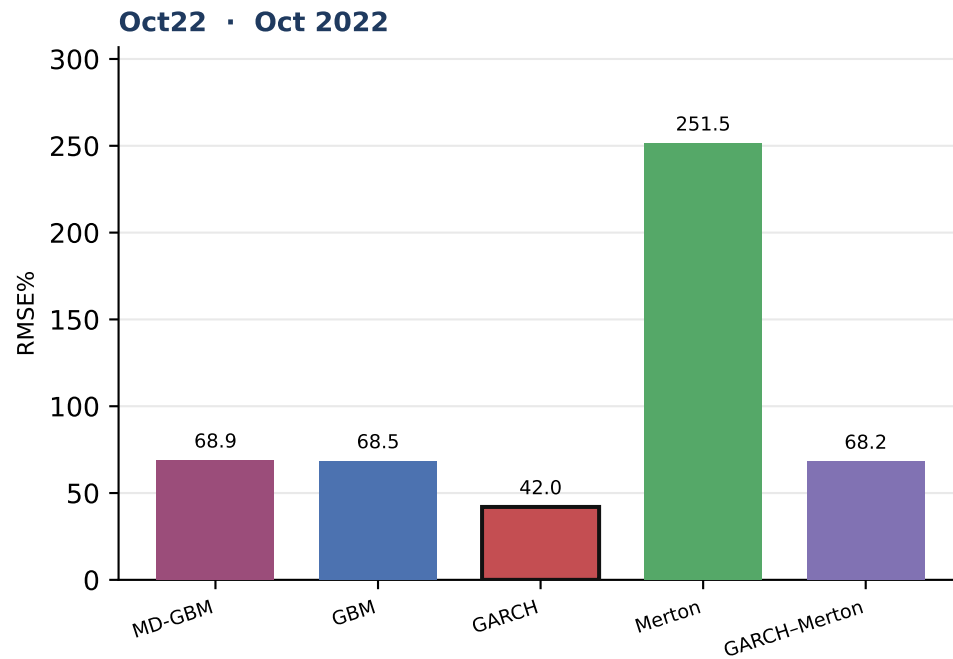
Model	Mean RMSE%	Median # windows best	
MD-GBM	92.51	98.55	0
GBM	92.13	98.04	1
GARCH	80.40	81.42	10
Merton	241.09	243.38	0
GARCH-Merton	92.08	98.94	1

Table S4.3 · MSFT

Model	Mean RMSE%	Median # windows best	
MD-GBM	83.87	97.81	0
GBM	83.61	97.51	2
GARCH	111.24	98.72	8
Merton	298.66	262.69	0
GARCH-Merton	84.10	98.22	2

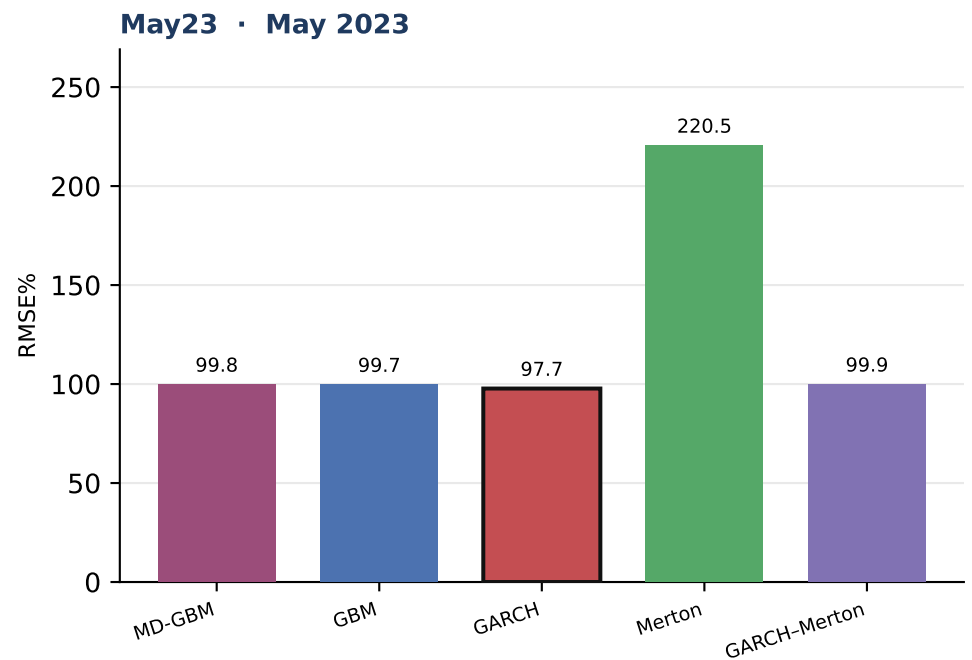
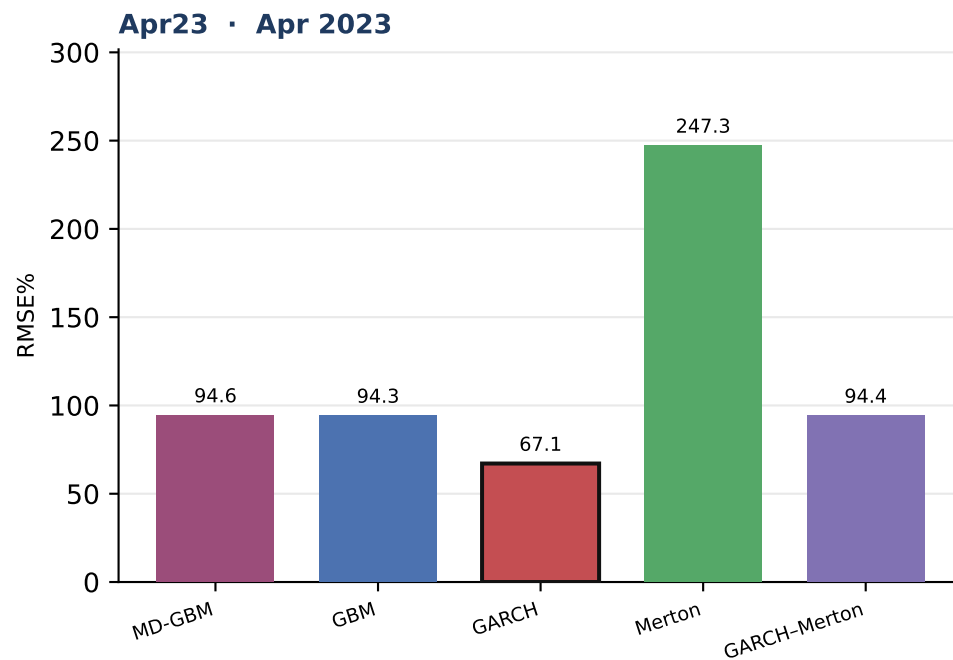
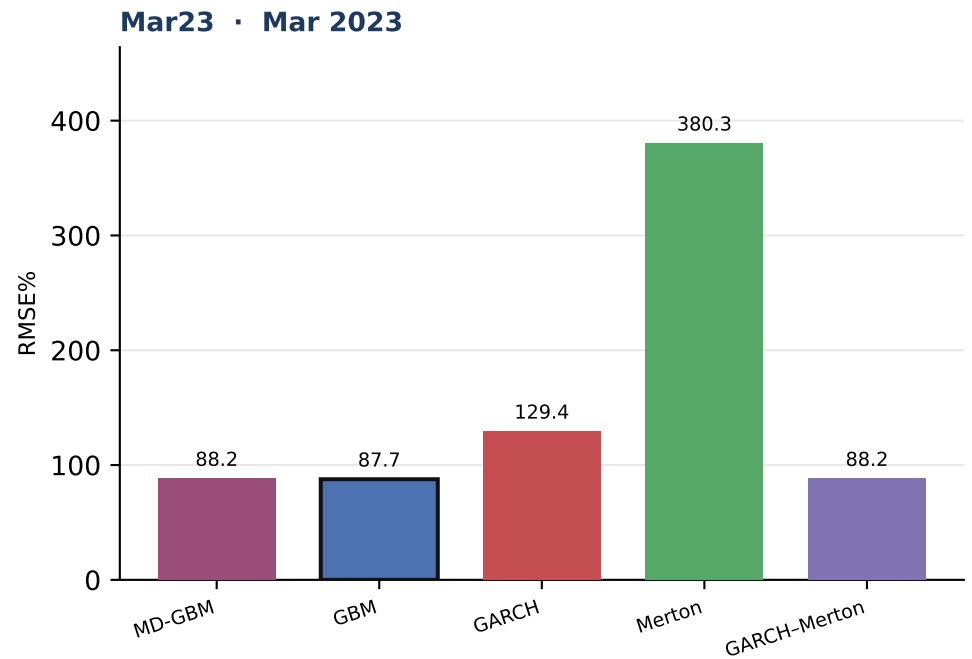
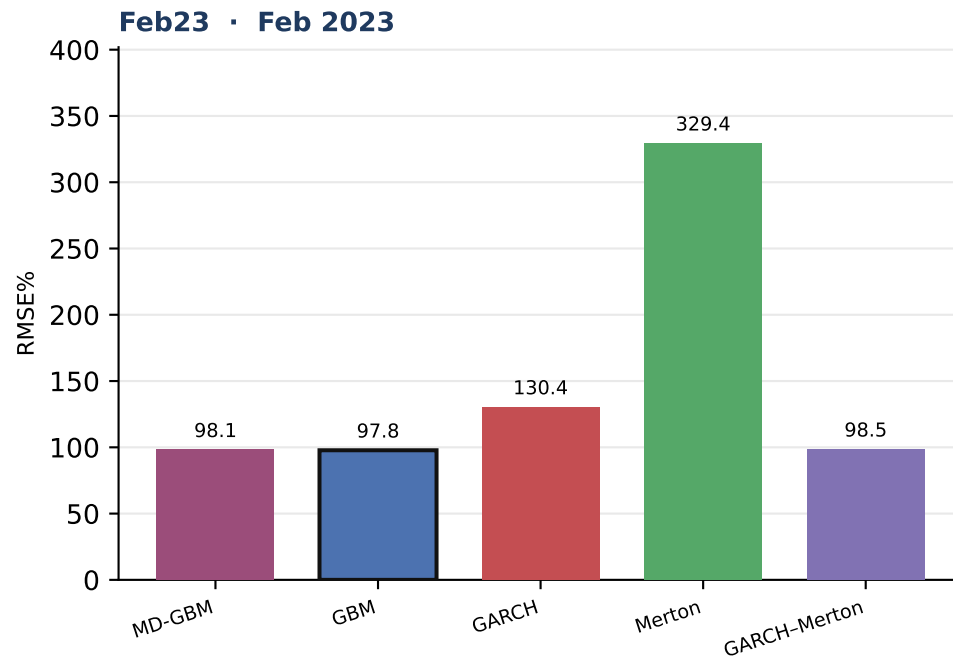
Each column is one company. Green row = lowest mean RMSE% across the twelve Friday-before-expiry windows for that name.

Figure 1 · RMSE% by model and window · mean of SPY / AAPL / MSFT



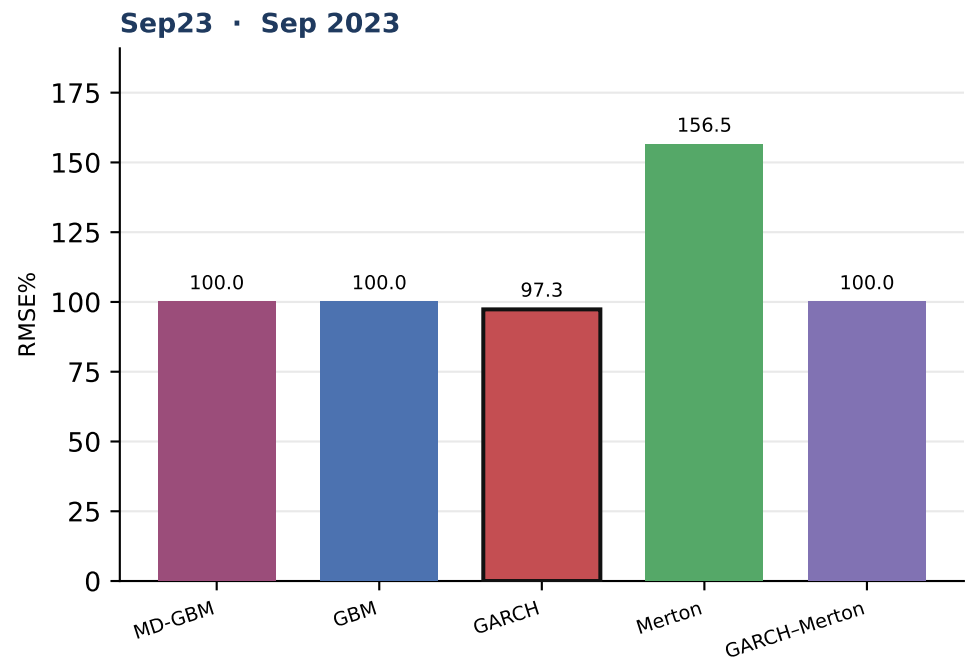
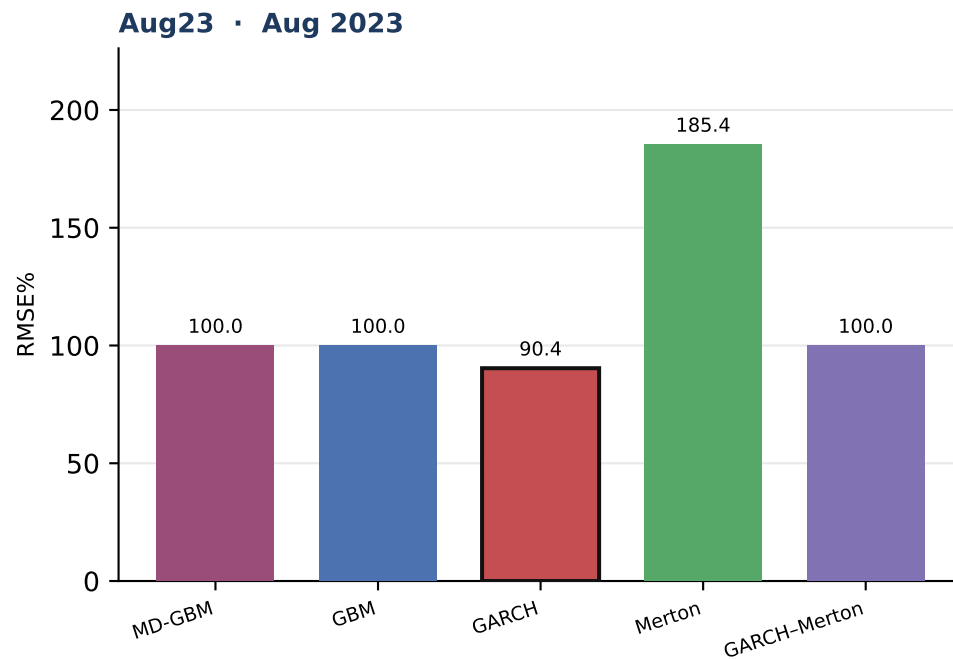
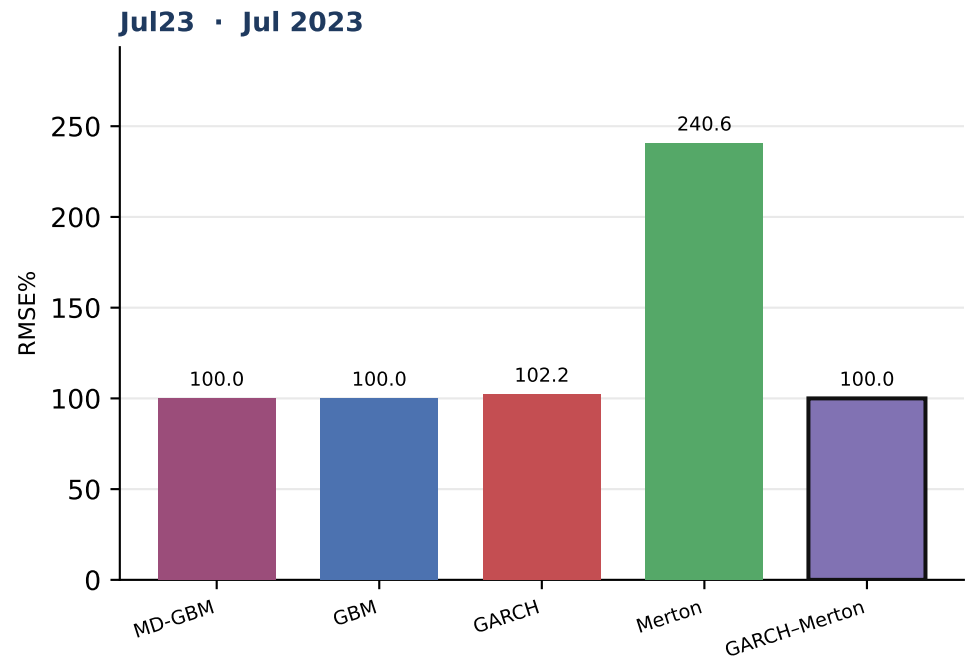
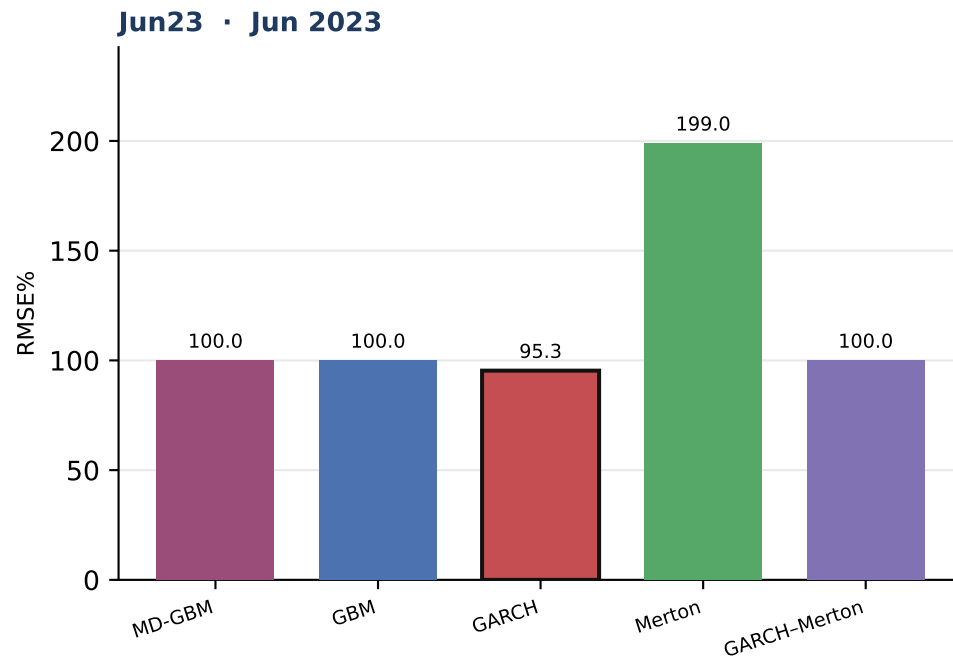
Outlined bar = lowest mean RMSE% in that window. Equal-weight mean of the three companies.

Figure 1 · RMSE% by model and window · mean of SPY / AAPL / MSFT



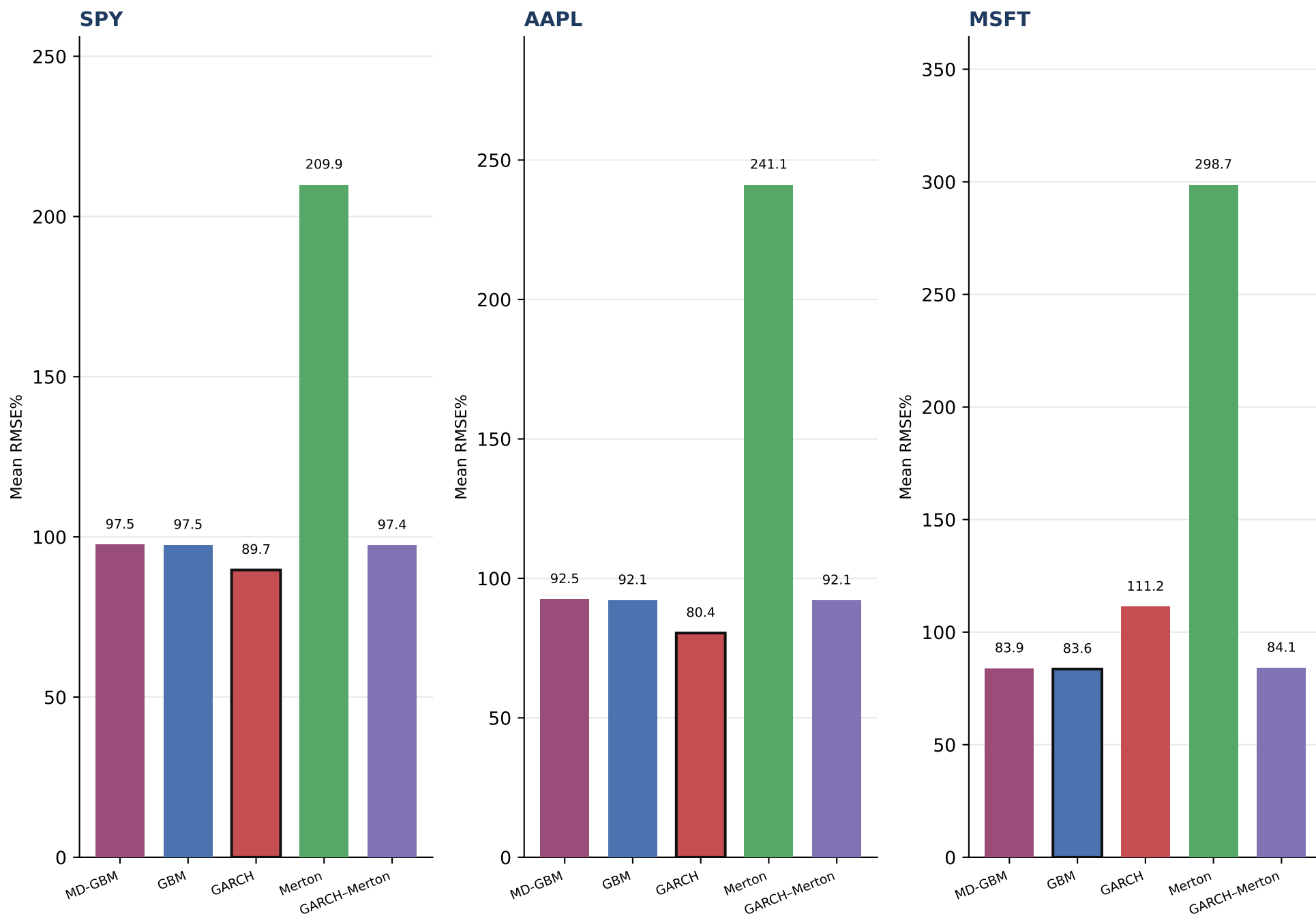
Outlined bar = lowest mean RMSE% in that window. Equal-weight mean of the three companies.

Figure 1 · RMSE% by model and window · mean of SPY / AAPL / MSFT



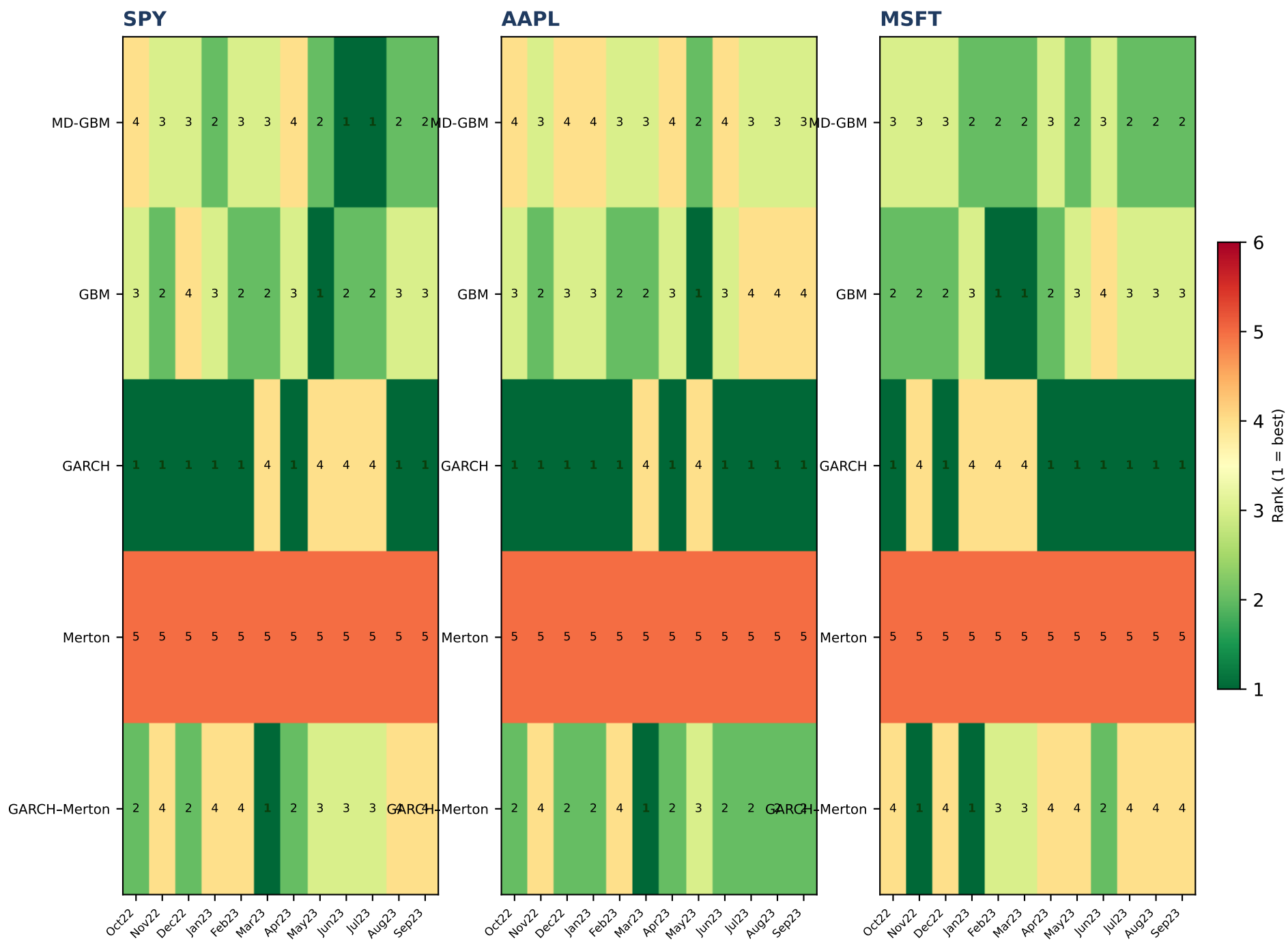
Outlined bar = lowest mean RMSE% in that window. Equal-weight mean of the three companies.

Figure 2 · RMSE% by model · each company (mean across twelve windows)



Outlined bar = lowest mean RMSE% for that company.

Figure 3 · Model rank by company × window (1 = best RMSE%)



Green / 1 = lowest RMSE% in that company×window cell. Rankings can and do change across names and windows.

Final analysis

Which models perform best, and does the ranking change?

- On mean percentage RMSE across the 36 company×window cells, GBM is the best overall model (mean RMSE% = 91.06; 4 of 36 cells).
- Second overall is GARCH-Merton (mean RMSE% = 91.18; 4 cells).
- The ranking is not stable. Best-in-cell models across the 36 tables: GARCH, GARCH-Merton, GBM, MD-GBM.
- SPY: lowest mean RMSE% is GARCH (89.68); wins 8 of 12 windows.
- AAPL: lowest mean RMSE% is GARCH (80.40); wins 10 of 12 windows.
- MSFT: lowest mean RMSE% is GBM (83.61); wins 2 of 12 windows.
- No-jump models as a group have lower mean RMSE% (92.42) than the three jump models (170.53).
- These conclusions are conditional on the V3 filters, 5-day hourly calibration, nearest-ATM hourly sample, LSM with 10000 paths and 5-minute steps to Friday 15:59, and American-call quotes only. They are not a statement about European options or other tenors.

How to read the 36 detailed tables

- Each of Tables SPY-1...MSFT-12 uses exactly the same option contracts for all six models. Differences are the dynamics, not the sample.
- Mark 1 is always lowest RMSE%. MAE, Bias \$, Bias%, and early-exercise are reported, not used for the ranking.
- A model that is Mark 1 on RMSE% can still be biased (systematically expensive or cheap).
- Early-exercise fractions are a model implication, not an accuracy score; LSM stops at window Friday close, not listed expiry.